

Selecting Ratio Representations for Comparisons

Answer Key

Grade 6

Quick Answers

- | | |
|--|--|
| 1. 15 | hours = 24, (c) who is ahead = > |
| 2. 8 | 8. (a) cents per yogurt in the pack = 75, (b) better buy = < |
| 3. 24 | 9. (a) cats at shelter A = 10, (b) more cats = > |
| 4. (a) Brand A = 3, (b) Brand B = 2, (c) cheaper brand = > | 10. cups of syrup = 12, — |
| 5. (a) Mia in 24 minutes = 60, (b) Noah in 24 minutes = 54, (c) faster reader = > | 11. (a) express train speed = 70, (b) local train speed = 70, (c) comparison = = |
| 6. (a) larger share, team A = 36, (b) larger share, team B = 35, (c) bigger of the two = > | 12. (a) stronger recipe = >, (b) cups of lemon juice = 16, — |
| 7. (a) hiker A in 6 hours = 27, (b) hiker B in 6 | |
-

What is verified

10 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6

6.RP.A.2 – *problems 1, 4, 8, 11*

6.RP.A.3 – *problems 2, 3, 5, 6, 7, 9, 10, 12*

Difficulty 1–5 of 5 across 27 tagged checks.

1. 6 pencils cost 90 cents. Cents per pencil?

Solution: Unit rate: $90 \div 6 = 15$ cents per pencil.

15

2. Cocoa to milk is 2 to 5. How much cocoa for 20 cups of milk?

Solution: A ratio table works well: the milk column goes from 5 to 20, a multiplier of

4. Cocoa scales the same way: $2 \times 4 = 8$ cups.

8

3. Split 42 cards in the ratio 3:4; find the larger share.

Solution: Tape diagram: $3 + 4 = 7$ boxes hold 42 cards, so one box is $42 \div 7 = 6$ cards. The larger share has 4 boxes: $4 \times 6 = 24$ cards.

24

4. Brand A: 4 lb for 12 dollars. Brand B: 5 lb for 10 dollars. Compare per pound.

Solution: (a) Brand A: $12 \div 4 = 3$ dollars per pound.

3

(b) Brand B: $10 \div 5 = 2$ dollars per pound.

2

(c) Comparing unit prices, $3 > 2$, so Brand B is cheaper per pound.

$3 > 2$, Brand B

5. Mia: 20 pages in 8 minutes. Noah: 27 pages in 12 minutes. Compare over 24 minutes.

Solution: 24 minutes is a common amount for both readers.

(a) Mia: $24 \div 8 = 3$ blocks of 8 minutes, so $20 \times 3 = 60$ pages.

60

(b) Noah: $24 \div 12 = 2$ blocks, so $27 \times 2 = 54$ pages.

54

(c) Over the same 24 minutes Mia reads more, so Mia is faster.

$60 > 54$, Mia

6. Team A splits 60 dollars in 2:3; team B splits 63 dollars in 4:5. Compare the larger shares.

Solution: (a) Team A: $2 + 3 = 5$ parts, one part = $60 \div 5 = 12$, larger share = $3 \times 12 = 36$ dollars.

36

(b) Team B: $4 + 5 = 9$ parts, one part = $63 \div 9 = 7$, larger share = $5 \times 7 = 35$ dollars.

35

(c) $36 > 35$: team A's larger share wins by a dollar.

$36 > 35$, team A

7. Hiker A: 9 km per 2 h. Hiker B: 12 km per 3 h. Compare after 6 hours.

Solution: (a) Hiker A: $6 \div 2 = 3$ blocks, $9 \times 3 = 27$ km.

27

(b) Hiker B: $6 \div 3 = 2$ blocks, $12 \times 2 = 24$ km.

24

(c) $27 > 24$, so hiker A is ahead.

$27 > 24$, hiker A

8. 8 yogurts for 600 cents, or one yogurt for 80 cents. Which is the better buy?

Solution: (a) Pack: $600 \div 8 = 75$ cents per yogurt.

75

(b) $75 < 80$, so the pack is the better buy.

$75 < 80$, the pack

9. Shelter A: cats to dogs 5:3, 16 animals. Shelter B: 9 cats. Which has more cats?

Solution: (a) Shelter A: $5 + 3 = 8$ parts for 16 animals, one part = 2, cats = $5 \times 2 = 10$.

10

(b) $10 > 9$, so shelter A has more cats.

$10 > 9$, shelter A

10. Syrup to soda is 3:5; the bowl holds 32 cups. Cups of syrup? Choose and defend a representation.

Solution: With a tape diagram: $3 + 5 = 8$ boxes hold 32 cups, one box $= 32 \div 8 = 4$ cups, syrup $= 3 \times 4 = 12$ cups. 12

Representation choice (instructor judges): Any of the three tools earns credit with a matching reason — e.g. a tape diagram because the 32 cups is a total split into ratio parts; a table or double number line scaling (3, 5) up to a syrup-plus-soda total of 32 also works.

11. Express: 210 km in 3 h. Local: 140 km in 2 h. Compare the speeds.

Solution: (a) Express: $210 \div 3 = 70$ km per hour. 70

(b) Local: $140 \div 2 = 70$ km per hour. 70

(c) The unit rates are equal — the bigger distance does not mean a faster train, because the times differ too. 70 = 70

12. Recipe A: juice to water 2:3. Recipe B: 3:5. Which is stronger, and how much juice for 40 cups of recipe A?

Solution: (a) Compare juice-per-water fractions: $\frac{2}{3} \approx 0.667$ and $\frac{3}{5} = 0.6$, so recipe A is stronger. $\frac{2}{3} > \frac{3}{5}$, recipe A

(b) Tape diagram for recipe A: $2 + 3 = 5$ boxes hold 40 cups, one box $= 40 \div 5 = 8$ cups, juice $= 2 \times 8 = 16$ cups. 16

(c) **Explanation (instructor judges):** Full credit names a comparison tool for part (a) — unit rates or the fractions — and a total-splitting tool for part (b) — tape diagram, table, or double number line — with reasons matching how each was used.